The ML Lifecycle

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

The ML Lifecycle

> Prepare The ML Lifecycle: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Distinguish **fitting** a rule from applying a fixed rule

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Creator Defense

Without opening the old lesson, explain one key idea from Creator Defense. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of generative systems defense

Reference example, not the only acceptable wording: Teacher explanation: A generative-system defense presents the question, architecture, corpus, permissions, baseline, **evaluation** and unresolved failures.

Demonstration fluency is only one observable property. Link each claim to a recorded result. State the distinction between retrieval relevance, claim support, fallback and permissions.

Give a reproduction packet and answer challenges without inventing missing measurements.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A team **label**s photos, fits a model and celebrates one correct new photo. List the missing **evaluation** steps and define the intended use before testing.

Concept to explain

Machine learning is an **evidence** cycle: define a **prediction** problem, prepare examples, fit a model, evaluate on unseen data, and improve one diagnosed cause.

Term	Meaning
machine learning	fit a model from examples
training	adjust model parameters using examples
inference	apply the fitted model to new input
evaluation	measure behavior on unseen examples

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach this Grade 9 **machine learning** lesson using the compact two-session route, or use the full three-session weekly plan. The central distinction is selection from **label**led **training** examples versus **evaluation** of a fixed rule on separate examples.

This is a deliberately small supervised classification exercise, not a deployed card recognition system. Students can complete it on paper; Python checks their trace. Do not add neural-network terminology or external accounts.

The **training** loop reports candidate mistakes, and students explicitly select and record the winner.

Readiness answer

3 of 4, or 75 percent, are correct in these cases. It does not prove correctness on future cases.

Printed lesson and workbook; six fictional worked cards

Paper to cover the held out rows until a threshold is recorded

Optional local Python 3 editor; no packages or personal data needed

Prepare a fresh retry card set from the assessment guidance

End goal for the pictorial case

By the end, I can explain the ml lifecycle using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model The ML Lifecycle using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and **evaluation** proof separately.

Question to explain

QUESTION Trace problem → data → train → predict → evaluate → improve, and identify where human judgment enters.

MODEL IDEA **Machine learning** is an **evidence** cycle: define a **prediction** problem, prepare examples, fit a model, evaluate on unseen data, and improve one diagnosed cause.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Goal: classify a leaf as healthy or spotted for a classroom plant helper. Prepare **label**ed leaf examples and reserve unseen test examples.

Draw a fresh data → model → **prediction** → **evaluation** trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Goal: classify a leaf as healthy or spotted for a classroom plant helper.

Step 2

STEP 2 Prepare **label**ed leaf examples and reserve unseen test examples.

Step 3

STEP 3 Fit the model on **training** data; predict **label**s for the test set.

Step 4

STEP 4 Measure errors, diagnose a cause, change one factor, and retest.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

> ### The teaching case
>
> A team **label**s leaf photos, trains a healthy/spotted classifier and shows one correct **prediction**.

> ### The teaching case
>
> Order the lifecycle and identify missing **evidence**.

Exact explanation to model

> ### Why the result follows
>
> Define use and **label**s, collect data, separate **evaluation** data, train, predict, evaluate and improve. One correct result is insufficient. Test unseen varied cases and retain human responsibility.

Explain the added worked example

Explain the added worked example · 1

In Grade 7 you programmed an activity chooser using a rule supplied by a person. This week you will choose a rule by comparing its **prediction**s with **label**led examples.

This small supervised learning task makes the learning **process** visible: people define the task and possible rules; **training** data determines which candidate fits best.

Explain the added worked example · 2

Imagine fictional practice cards with a measured width x and a category **label**, 0 or 1. Width uses invented units. We want to predict the category of another card from its width.

The **label**s are provided **observation**s in this exercise, not values generated by our **prediction** code. These invented cards demonstrate a method; they provide no **evidence** about real objects or students.

Explain the added worked example · 3

Before **fitting**, set aside the final two cards. They are visible here so you can follow the complete example, but do not use them to choose a rule. Our candidates predict 1 when width is at least 2, 4 or 6, and 0 otherwise.

Choose the candidate with the fewest **training** mistakes. If candidates tie, choose the smaller threshold; fix this rule before testing.

Explain the added worked example · 4

int(True) is 1 and int(False) is 0. For threshold 2, the **training prediction**s are 0, 1, 1, 1: the width 3 card is wrong. For threshold 4, they are 0, 0, 1, 1: none is wrong.

For threshold 6, they are 0, 0, 0, 1: the width 5 card is wrong. The printed rows are 2 1, then 4 0, then 6 1. Select threshold 4 using this **training evidence** and record it.

The loop reports errors; the learner makes and records the selection.

Key terms

Term	Meaning
Feature	An input value used to make a prediction
Label	An observed target value supplied for an example
Training	Using examples to select or adjust a model
Held out data	Examples reserved from fitting and model choices for evaluation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 5

Freeze means do not change the chosen threshold during this **evaluation**. With threshold 4, widths 2 and 6 receive **prediction**s 0 and 1. Compare them with the held out **label**s: both match. Report 2 correct out of 2 held out cards.

Training results and test results answer different questions: did the rule fit the examples used to select it, and how did that fixed rule perform on these separate examples?

> **Keep threshold 4 fixed. An additional card has width 4 and observed label 0. Predict its category, decide whether it is correct, and update the held out count from 2 of 2.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: The ML Lifecycle

Follow one complete plant-classifier lifecycle without tuning on the final test.

The facts we will use

> **Supplied facts**
>
> Toy **training** leaves have widths 1 healthy, 3 healthy, 5 spotted, 7 spotted. Candidates predict spotted at width>=2 or >=4.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/1

The ML Lifecycle

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fit using training

Cutoff 2 makes one error at width 3; cutoff 4 makes zero training errors.

2. Freeze candidate

Choose cutoff 4 under the stated training-error rule.

3. Test once

Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

CHANGE ONE THING

The fresh width-6 leaf is actually healthy.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit using training	Cutoff 2 makes one error at width 3; cutoff 4 makes zero training errors.
2. Freeze candidate	Choose cutoff 4 under the stated training-error rule.
3. Test once	Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Two invented test leaves cannot establish useful real-world plant classification.

Change one thing in the pictured case

Change one thing

Change: The fresh width-6 leaf is actually healthy.

Predicted result

Prediction: The frozen model is correct on 1/2 test leaves.

Why the result follows

Reason: The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Ask while showing the picture

Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate The ML Lifecycle, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Board sequence

1. Put the facts on the board

Write the given facts: A team **label**s leaf photos, trains a healthy/spotted classifier and shows one correct **prediction**.

2. State the decision to explain

Write the question to explain: Order the lifecycle and identify missing **evidence**.

3. Reveal the worked reasoning

Reveal after **prediction**s: Define use and **label**s, collect data, separate **evaluation** data, train, predict, evaluate and improve. One correct result is insufficient. Test unseen varied cases and retain human responsibility.

Hold back the independent task

Keep for an independent attempt: A team selects a model before deciding what outcome to predict or where the data comes from. Name two missing lifecycle decisions and their effect.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The prerequisite answer is 3 of 4, or 75 percent, for those four cases only. Guided case: width 4 produces 1 at threshold 4, but its **label** is 0, so it is wrong; the updated held out result is 2 of 3.

Exit: a supported claim is that the fixed threshold matched both of these two cards. It does not show all future cards will be correct.

A reasonable next step is to evaluate more unused examples chosen to represent the intended task, while recording the sampling limits.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the Grade 7 idea of testing a fixed rule. Ask the prerequisite question and require a denominator in the answer.
7 to 18 minutes	Define width, label and threshold. Read the card table. Cover the held out rows and agree the three candidate values and tie rule before fitting.
18 to 32 minutes	Trace every prediction for thresholds 2, 4 and 6. Use W1 and W2. Have students explain a wrong prediction rather than only counting errors. Optionally run the Python loop.
32 to 45 minutes	Record threshold 4, then uncover held out cards. Complete the guided additional width 4 case. Compare training and held out results with separate denominators.
Session 2 start to 8 minutes	Retrieve the sequence define, separate, fit, freeze and test. Ask which step may use training labels and which records test results.
8 to 20 minutes	Discuss the unfair evaluation and complete W3. Show why changing a model after seeing test labels changes the status of those examples.
20 to 35 minutes	Collect W4 independently. Have each learner record the training errors and threshold before revealing the test rows. Record whether support was needed.
35 to 45 minutes	Discuss W5 and the exit claim. Offer the fresh retry after support where needed. Preserve both the original result and later correction.

Reduce help deliberately

Completion practice

Completion task: Use threshold 6 on the four **training** widths 1, 3, 5 and 7. Write all **prediction**s, identify each error, and explain why threshold 4 is selected.

Reasoning to listen for

Teacher check: **Prediction**s are 0, 0, 0, 1. The width 5 card is wrong because its **label** is 1. Threshold 6 has one **training** mistake; threshold 4 has zero and is selected.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for The **machine learning** lifecycle. Proposed explanation: “The model wrote its own task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: People chose the **input**, **label** meaning, candidate thresholds and selection rule. Learning here means choosing a parameter using the **training** examples.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

No. Report 1/2 for the frozen model. Any revision needs development data and a newly reserved final **evaluation**. Changed-case check: The frozen model is correct on 1/2 test leaves.

The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Changed-case reasoning: The frozen model is correct on 1/2 test leaves. The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the The ML Lifecycle investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Lifecycle Relay: place twelve action cards in order, justify every handoff, then repair two broken workflows.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR **Training** once and seeing one correct **prediction** proves the model works.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The model wrote its own task: People chose the **input**, **label** meaning, candidate thresholds and selection rule. Learning here means choosing a parameter using the **training** examples.

Zero **training** mistakes means no future mistakes: **Training** performance measures the examples used to choose the threshold. The additional width 4 card shows how a fitted model can be wrong on a different example.

A better score makes reused test data independent: Once test **label**s influence a revision, the examples are part of development. Preserve the history and use unused cases for a later independent **evaluation**.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the The ML Lifecycle questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Define use and **label**s, collect data, separate **evaluation** data, train, predict, evaluate and improve. One correct result is insufficient. Test unseen varied cases and retain human responsibility.

Independent question

Independent question · withhold the key

A team selects a model before deciding what outcome to predict or where the data comes from. Name two missing lifecycle decisions and their effect.

Expected independent answer

Teacher key · Expected independent answer

Define the intended use and target, then inspect data suitability and **evaluation** criteria. Without these, a model score may answer the wrong problem.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Choosing an advanced model is always the first meaningful ML step.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Specify users, **label**s and operating conditions; separate related examples before development; compare a baseline; evaluate on varied untouched cases with declared metrics; document failures and human review.

One success supplies one **observation**, not broad reliability.

Explanations for the additional workbook examples

W1 Identify the learning decision

Workbook question · W1

For the worked cards, name the **input**, **label**, three candidate thresholds and the information allowed to select a threshold.

Teacher reasoning · W1

The **input** is width x and the **label** is the observed category 0 or 1. Candidates are 2, 4 and 6. Selection uses mistakes on the four **training** cards only; held out **label**s must not guide it.

W2 Trace one candidate

Workbook question · W2

Use threshold 6 on the four **training** widths 1, 3, 5 and 7. Write all **prediction**s, identify each error, and explain why threshold 4 is selected.

Teacher reasoning · W2

Predictions are 0, 0, 0, 1. The width 5 card is wrong because its **label** is 1. Threshold 6 has one **training** mistake; threshold 4 has zero and is selected.

W3 Repair the report

Workbook question · W3

A learner fits on all six worked cards and then reports 6 of 6 as independent test performance. Explain what is wrong and what **evidence** is needed.

Teacher reasoning · W3

All six cards influenced **fitting**, so none of those six provides an independent test. Report this as performance on **fitting** data.

Freeze the rule and evaluate it on unused examples; do not claim a new split of the same reused cards repairs the earlier **evaluation**.

W4 Fit and evaluate independently

Workbook question · W4

New **training** cards are (width 2, **label** 0), (4, 0), (6, 1), (8, 1). Candidates are 3, 5 and 7. Predict 1 when width is at least the threshold.

Count **training** mistakes, select the fewest with smaller threshold breaking ties, and record the choice before testing. Held out cards are (3, 0), (7, 1), (5, 0).

Give each test **prediction**, the correct count and a limited conclusion.

Teacher reasoning · W4

Training mistake counts for thresholds 3, 5 and 7 are 1, 0 and 1; select 5 before examining test results. Held out **prediction**s are 0, 1, 1. The width 5 card is wrong, giving 2 correct out of 3.

This supports the result on these three cards only, not universal reliability. Preserve threshold 5 for this **evaluation**.

W5 Decide the next step

Workbook question · W5

After W4, a team changes its threshold because of the test error. Can the same three cards be an untouched final test for the changed model? Explain how to proceed.

Teacher reasoning · W5

No. Their **label**s have influenced the revision. Keep the first result, log the new threshold and use the old cases for development or regression checks. Use fresh reserved cases for an independent final **evaluation**.

Assess independent transfer

Independent question · withhold the key

Selected case: New **training** cards are (width 2, **label** 0), (4, 0), (6, 1), (8, 1). Candidates are 3, 5 and 7. Predict 1 when width is at least the threshold.

Count **training** mistakes, select the fewest with smaller threshold breaking ties, and record the choice before testing. Held out cards are (3, 0), (7, 1), (5, 0).

Give each test **prediction**, the correct count and a limited conclusion.

Selected case answer

Teacher key · Selected case answer

Training mistake counts for thresholds 3, 5 and 7 are 1, 0 and 1; select 5 before examining test results. Held out **prediction**s are 0, 1, 1. The width 5 card is wrong, giving 2 correct out of 3.

This supports the result on these three cards only, not universal reliability. Preserve threshold 5 for this **evaluation**.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

TEACHER GUIDE · 6 OF 9

Connect and extend learning

> Connect The ML Lifecycle to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Write the Responsible Plant Classifier problem statement, intended user, **prediction** target, success measure, and human-review point.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny **label**led dataset, explicit units and a named positive class. Stretch: require a baseline, held-out **evaluation** and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 2: **Feature**s, **Label**s, Targets. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use a **prediction** grid with one row per card and one column per candidate. Cover held out rows until the choice is written. Read pairs aloud as width then **label**.

Extension: candidates 4 and 5 both fit the worked **training** set with zero errors; the predeclared smaller-threshold tie rule chooses 4. Explain why the test **label**s must not break this tie.

Require W4 **training** counts 1, 0, 1, selection 5, held out **prediction**s 0, 1, 1 and result 2 of 3, plus an explanation of separation and limits. Assess the explanation before offering hints.

Fresh retry: **training** (1,0), (5,0), (7,1), (9,1), candidates 4,6,8. Mistakes are 1,0,1; choose 6. Held out (2,0), (8,1), (6,0) produces 0,1,1 and 2 of 3. Ask why the final error must remain in the report.

Correct arithmetic without keeping **fitting** and testing separate is not sufficient exit **evidence**.

Week 2 examines **feature**s, **label**s and which information exists at **prediction** time. Carry forward the recorded split, threshold and test result.

Teacher reference for data separation and leakage: https://scikit-learn.org/stable/common_pitfalls.html#data-leakage (reviewed 2026-09-26).

Plan the delayed return

Suggested return points in this level: ai-4/2, ai-4/4, ai-4/7. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed The ML Lifecycle lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Distinguish **fitting** a rule from applying a fixed rule

Prepare the **label**led worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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Before class, solve the supported practice: Use threshold 6 on the four **training** widths 1, 3, 5 and 7. Write all **prediction**s, identify each error, and explain why threshold 4 is selected.

Supported practice answer

Reasoned practice answer: **Prediction**s are 0, 0, 0, 1. The width 5 card is wrong because its **label** is 1. Threshold 6 has one **training** mistake; threshold 4 has zero and is selected.

Check every learner before discussion: Use the worked case for The **machine learning** lifecycle. Proposed explanation: “The model wrote its own task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: People chose the **input**, **label** meaning, candidate thresholds and selection rule. Learning here means choosing a parameter using the **training** examples.

Support without giving away the answer

Support: read the exact case aloud, enlarge the **label**led picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The fresh width-6 leaf is actually healthy. Require a **prediction** and a reason before comparing. Expected result: The frozen model is correct on 1/2 test leaves.

Reason: The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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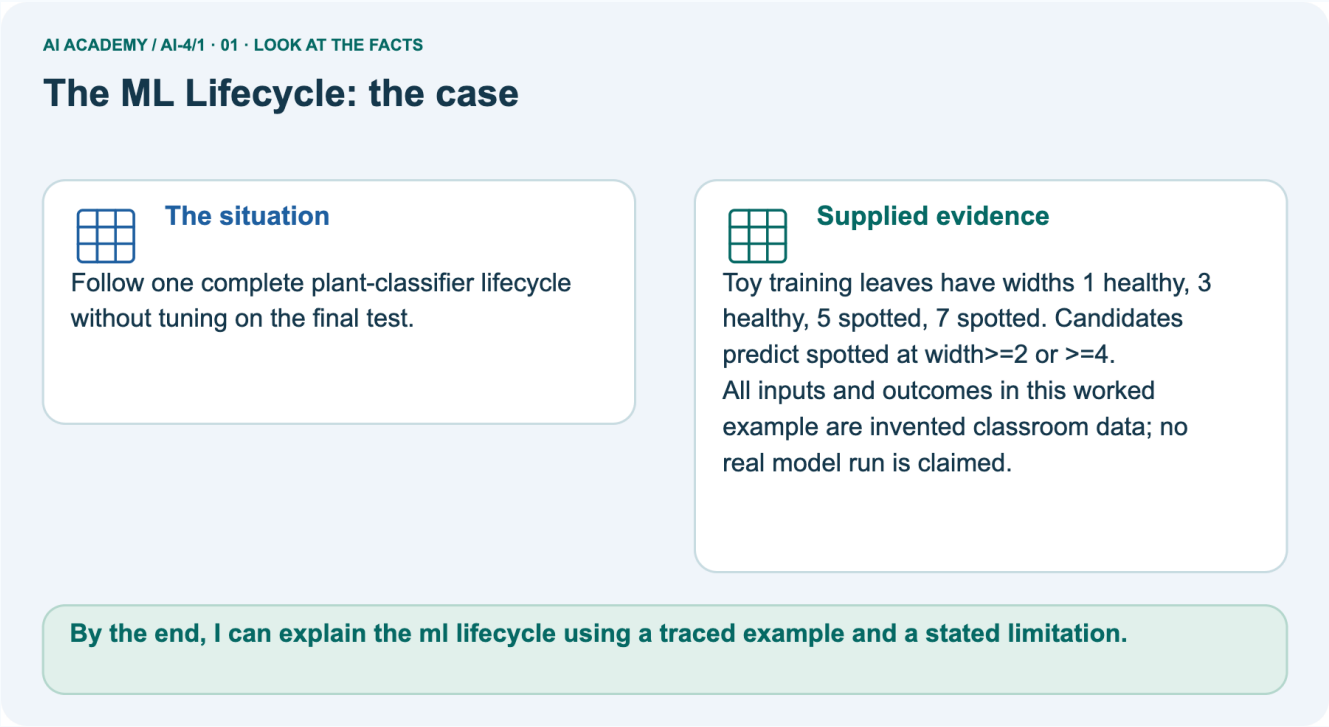
Visual concept journey

> Show one The ML Lifecycle graphic at a time, request a prediction and discuss its trace

> before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. The ML Lifecycle: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Follow one complete plant-classifier lifecycle without tuning on the final test. Supplied facts: Toy **training** leaves have widths 1 healthy, 3 healthy, 5 spotted, 7 spotted. Candidates predict spotted at width>=2 or >=4.

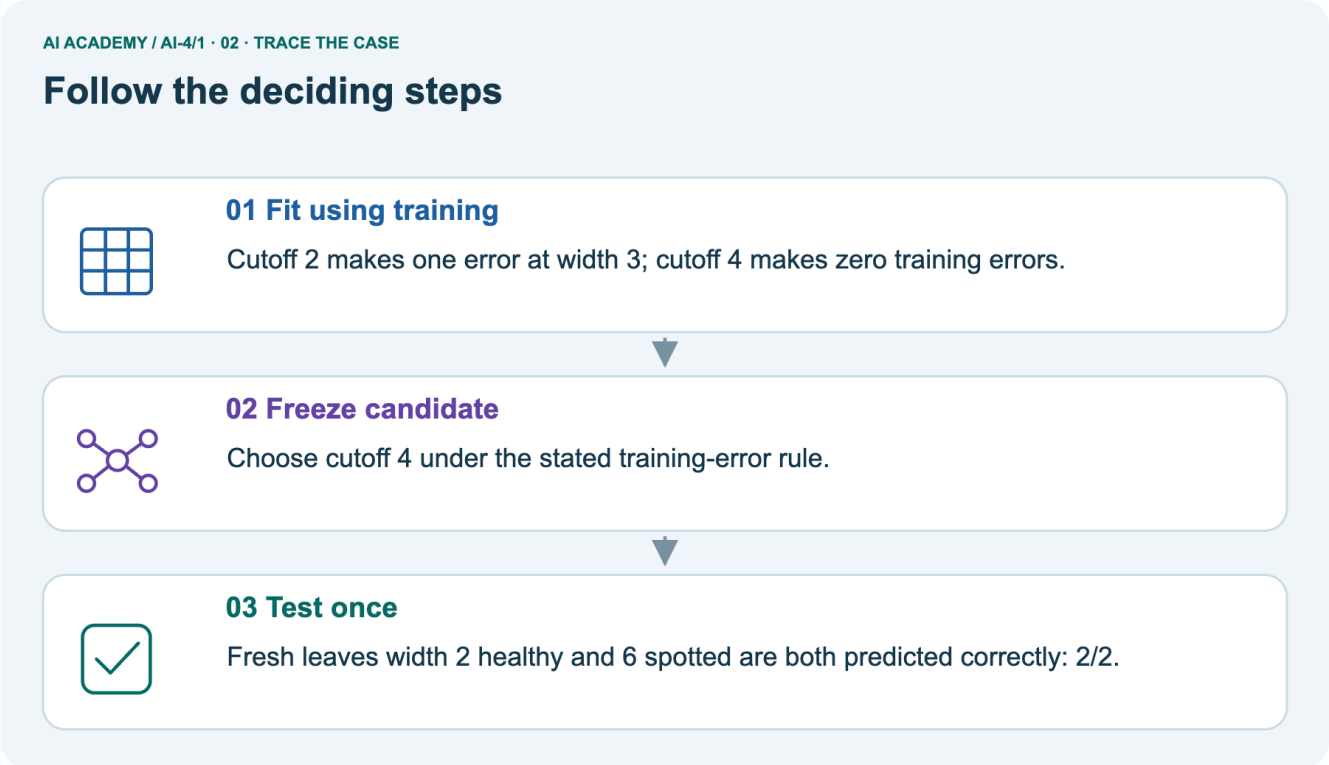
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

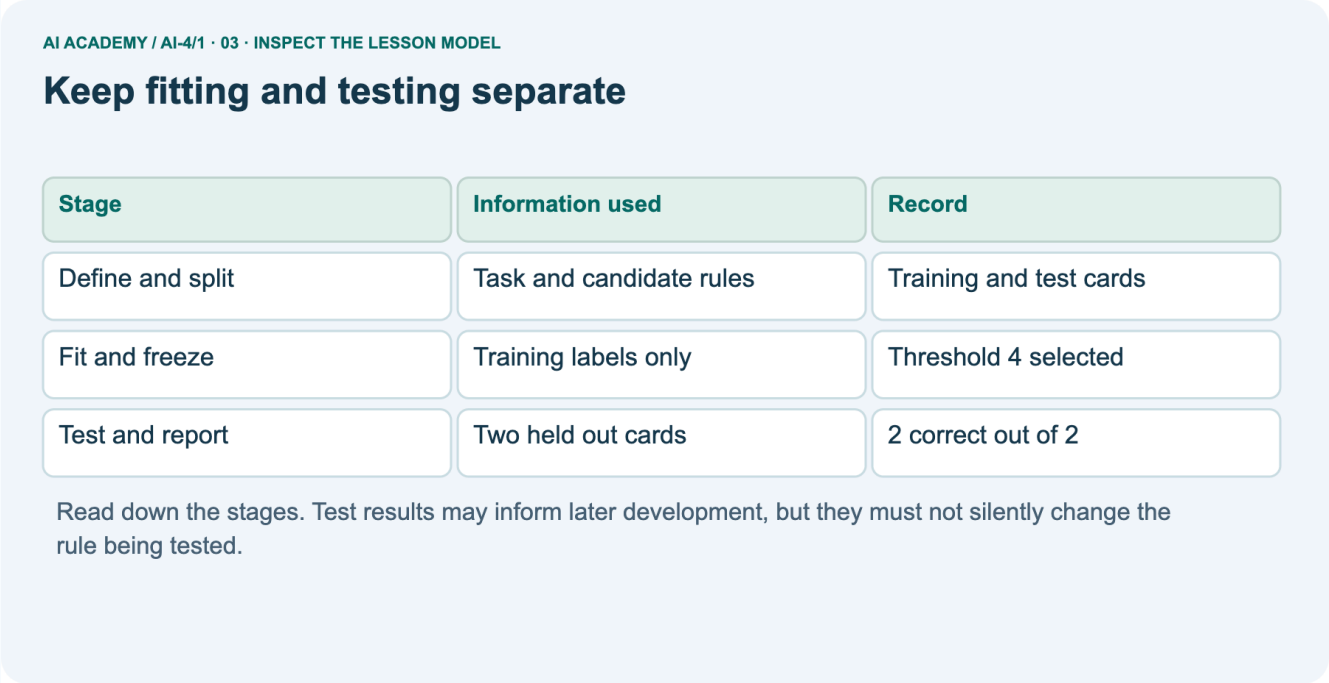
Fit using **training**: Cutoff 2 makes one error at width 3; cutoff 4 makes zero **training** errors. → Freeze candidate: Choose cutoff 4 under the stated **training**-error rule. → Test once: Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Keep fitting and testing separate



Follow the labelled picture with the matching explanation. Open the vector diagram

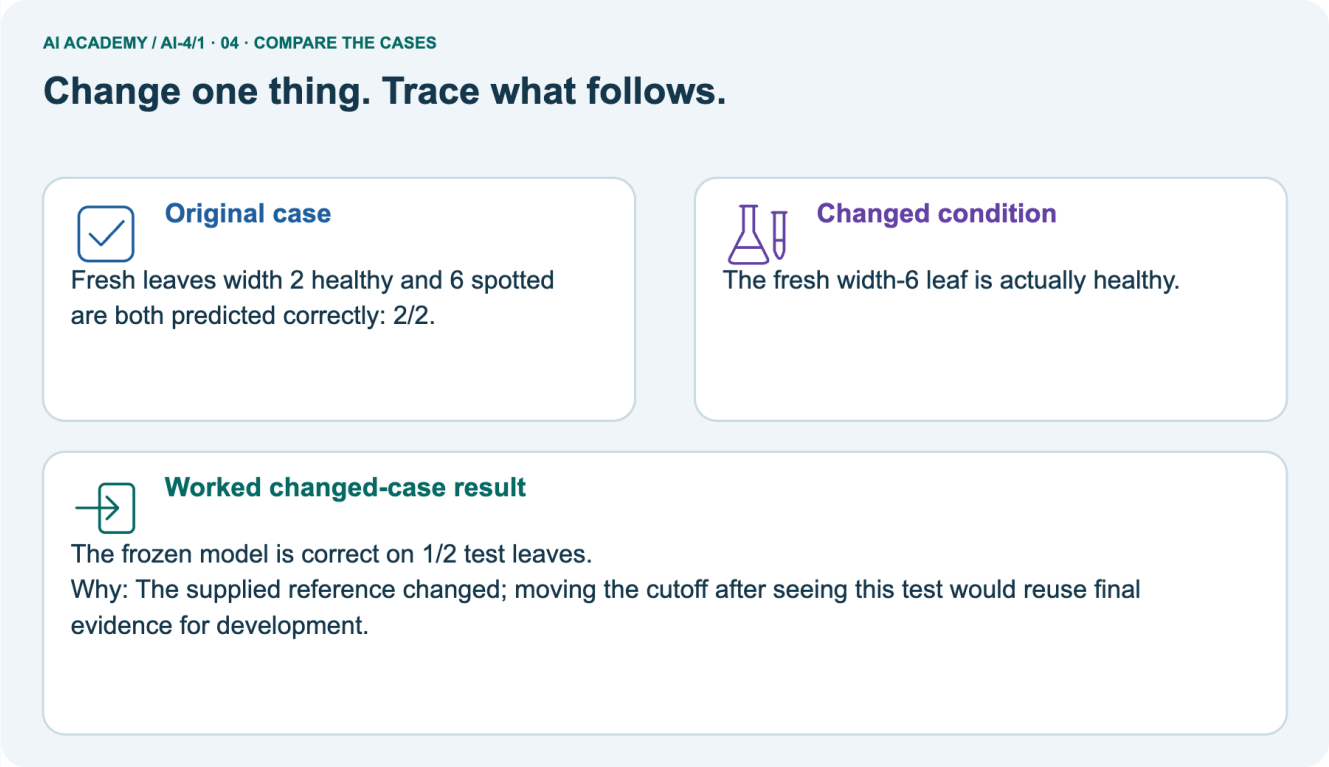
Keep **fitting** and testing separate. Define and split | Task and candidate rules | **Training** and test cards. Fit and freeze | **Training label**s only | Threshold 4 selected. Test and report | Two held out cards | 2 correct out of 2

Read the **label**s and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the **label**s and values. Explain the relationship the model shows. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. Changed condition: The fresh width-6 leaf is actually healthy. Expected result: The frozen model is correct on 1/2 test leaves.

Reason: The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

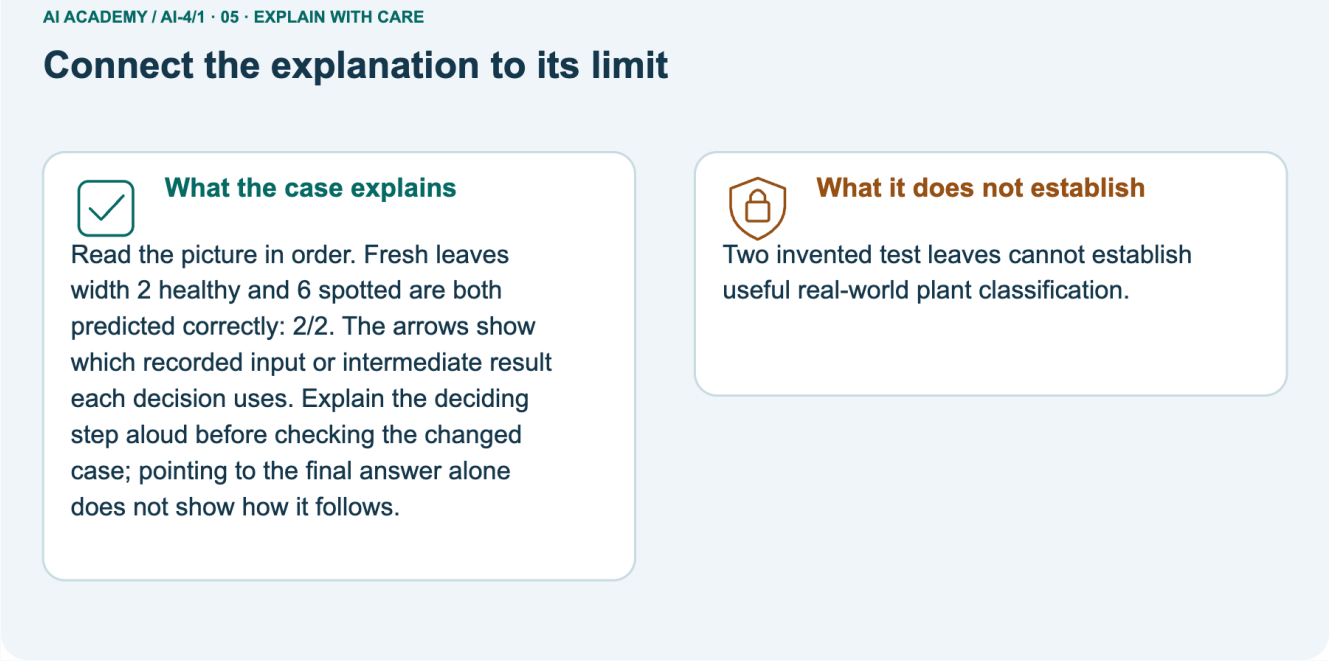
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one **label** or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Fresh leaves width 2 healthy and 6 spotted are both predicted correctly: 2/2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Two invented test leaves cannot establish useful real-world plant classification.

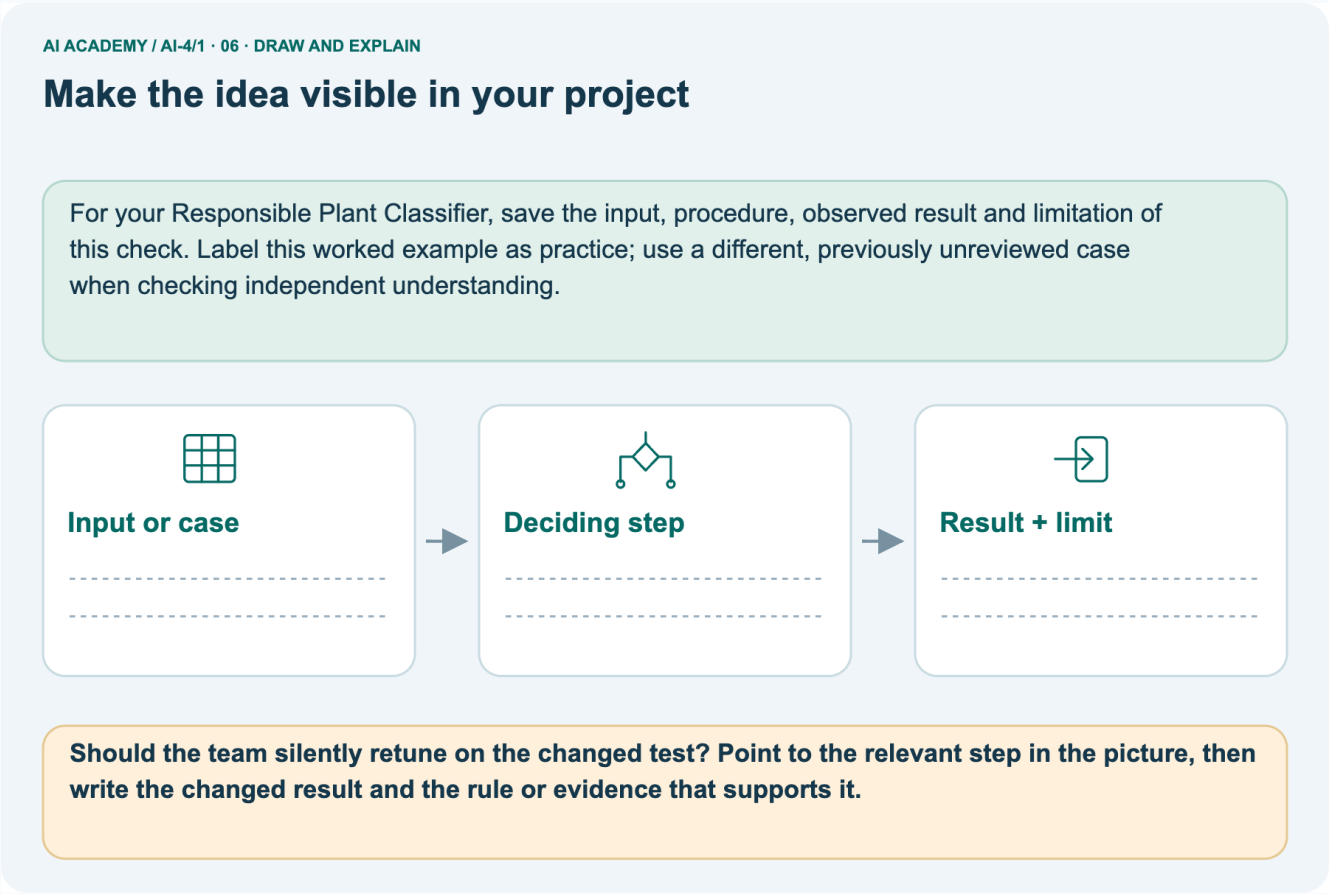
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Two invented test leaves cannot establish useful real-world plant classification.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Report 1/2 for the frozen model. Any revision needs development data and a newly reserved final **evaluation**. Changed-case check: The frozen model is correct on 1/2 test leaves.

The supplied reference changed; moving the cutoff after seeing this test would reuse final **evidence** for development.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.